

Considerations on Preserving VR Experiences (ideas collected over thirty years)

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This presentation delves into my personal reflections on preserving VR experiences over the course of my 30+ years in the field and look forward to how the community of VR can do more to ensure the beginnings of the field (and significant steps along the way) can be preserved. My father was a history professor, so even though my calling was, perhaps, in the completely opposite direction, I still acquired an appreciation for knowing about what came before, and the importance of knowing how people experienced life in the past. (Also, a practical need that drove some of my efforts to capture experiences was the need to present the possibilities of what VR offers to students in my classes, or even readers of my books.) Here we look at what archiving of VR have been accomplished thus far, what more we can do going forward, with other considerations about capturing immersive experiences.

Perhaps the more successful method of archiving VR experiences that we can consider historical is through video capture [3]. Many video captures were made because an application was developed and promoted during the so-called “Peak of Inflated Expectations” phase of the technology life cycle, and promo videos, B-roll footage, and news packages were shot and edited of these experiences. In some cases, experiences were recorded specifically for archival purposes (something the teams I’ve collaborated with have done).

Not all video capture is the same. Videos can be captured either (or both) from the POV of an external viewer, and from the first-person POV of the participant. For HMDs, the latter is easier, whereas for CAVEs, the former. But for CAVEs, it is also possible to capture the video feed to each of the display surfaces, but this too can have issues. Ideally, one could convert captures of the rendered surfaces of a CAVE into a 360 video – or even have a special external display dedicated to producing 360 output. The advantage of a 360-video capture is that viewers of the archived material would still have a modicum of interactivity through the ability to look around in any direction they choose.

An alternative to video archival is the written word, side-by-side with pictures. Fortunately, many applications are captured this way as the experience creators wish to publish their work, which may be ancillary to some experimental study, or be part of an interesting system construction. Thus, we still have access to reasonable descriptions of early applications through research publications. Other publications are broader and were written to exemplify the wide scope of possibilities that VR (and AR) provide(d). Specifically in my case, the books that I co-authored made sure to include as many examples as possible of real VR experiences when expounding on a concept. In one case (*Developing VR Applications* [2]), my co-author and I worked hard to best capture nearly 50 VR experiences through interviews with the creators as well as reading their writings, reviewing any video material, and, when pos-

sible, personally interacted with the experience. The intent of that effort was to provide exemplary examples of VR in a variety of fields from which new VR creators could be inspired. Today it also serves as an archive of the VR of the 1990’s. In the latest edition of our textbook (*Understanding Virtual Reality* [5]), we even included a separate index of media experiences.

Of course, probably the best way to “archive” an experience is to preserve the code and data that enable the experience to be executed on a computer and keep the experience alive in the most direct way. One issue that arises is that things decay: tools change, hardware changes (dies and replaced by something new), and operating systems evolve, all of which can cause the functionality of the code to falter. So perhaps we need to distinguish between operational code and non-operational code. And maybe sometimes we can move an experience from the latter category to the former.

Examples where the loss of requisite software has impeded the continued execution of an experience are the Iris Performer library from SGI, as well as the IrisGL precursor to OpenGL. In the latter case, porting from IrisGL to OpenGL is a fairly low hurdle, requiring minimal motivation to make it happen (and something I have done for a handful of early CAVE applications such as “*Detour: Brain Deconstruction Ahead*” [1], EVL’s “*Earth*”, “*Sandbox*”, and others. Performer-based code, however, is a larger hurdle, and I am not aware of any applications that have been ported (though I have it on good authority that a few SGI systems connected to CAVE displays are still operational). Likewise, many of the early CAVE applications were written using the EVL CAVE library, which itself is still commercially available, but also, the API of the FreeVR [4] library makes for an easy port from CAVElib. This talk will also consider other implications of archiving experiences through code – especially how porting code to modern API’s may raise some interesting questions, like: can I make other changes to the code while porting? Is the act of porting already changing the nature of the experience? And others.

Finally, we consider: “What more we can do” to archive VR. Primarily what we can do is to at least make video captures of participants in the experience, and then make good archives of the code and data. As part of that, we might consider a mechanism by which all this can be archived, which to some degree Github does for modern projects.

I believe some great VR experiences have already been lost to the ages. My favorite lost experience is “*Nexus Plexus*”, an Escher-like world where gravity changes as the participant jumps from object to object – an application I had the pleasure of experiencing firsthand at the Iowa State 6-sided CAVE nearly 25 years ago. Let’s strive to avoid the loss of future great experiences.

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William Sherman is a Computer Scientist for the National Institute of Standards and Technology (NIST) in the High-Performance Computing and Visualization Group. He is interested in all types of immersive technology, scientific visualization methodologies, as well as the merging of the two.

Prior to joining NIST, William worked on visualizations, both immersive and non-immersive for the Indiana University Advanced Visualization Lab. Earlier, at the Desert Research Institute (DRI),

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he established the Center for Advanced Visualization, Computation, and Modeling which housed both a 4-sided, and a 6-sided CAVE. At the National Center for Supercomputing Applications (NCSA), he led the technical efforts of the VR lab starting in 1993. In 1994 the NCSA VR lab constructed CAVE #2 with the assistance of the EVL team at the University of Illinois, Chicago. He has been working in VR for over 30 years.

William has also taught courses on virtual reality and scientific visualization to undergrad and graduate students for the University of Illinois at Urbana-Champaign, the University of Nevada-Reno, and Indiana University. (The VR courses used the prevailing technology of the time – and thus what students had access to – from CAVE systems to Google Cardboard to HTC-Vive HMDs.) All along the way, Sherman has been doing what he can to capture and convey what has been accomplished in VR, as well as what else might be possible. Sherman is also the co-author or editor of four books on virtual reality.

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